

their outputs. Stanford University's research into AI over-reliance, particularly in high-stakes fields like healthcare and law, underscores the dangers of accepting AI decisions without scrutiny. The study advocates for simpler AI explanations and more accountable decision-making processes, highlighting the need for tools that users can understand and trust. This scenario exemplifies the risks associated with proprietary systems' lack of transparency, emphasising the importance of open source models that offer greater inspectability and modifiability.

The innovation and sustainability bottleneck

Exclusively focusing on proprietary models not only stifles innovation by limiting access to underlying technologies but also raises significant environmental concerns. The development of open source LLMs fosters a culture of collaborative innovation, where a diverse community contributes to advancements and solutions. This approach not only accelerates technological progress but also distributes the developmental load, reducing the carbon footprint associated with creating large models from the ground up.

Customisation and IP concerns

Before going into the depth of risks around customisation and IP, let's try to understand the need with a scenario. In the world of generative AI applications, the debate often centres on whether to develop custom solutions for specific use cases. Use cases are endless, which means creation of custom applications for every use case may lead to a huge number of applications running around the organisation. Can we then take an approach that OpenAI took? They provided a chatbox and said, do whatever you want to do! We all then learnt different prompting techniques to use this in our world.

While connections to structured databases or external knowledge bases may necessitate unique applications, many needs can be met through the adept use of existing LLMs as is. The crux lies in crafting effective prompts to leverage the model's inherent knowledge base for tasks like generating organisation-specific documents or marketing content.

However, apprehensions about intellectual property (IP) and data privacy significantly deter organisations from utilising platforms like ChatGPT, even in its enterprise variants. The reluctance stems from concerns over data security and IP confidentiality, despite assurances from service providers about data protection. This scenario underscores a critical advantage of open source LLMs: the ability to host these models within an organisation's own infrastructure, offering complete control over data and IP without compromising on the capabilities offered by generative AI.

Imagine creating a marketing campaign or drafting emails with a tool that mirrors ChatGPT's functionality but operates entirely within your organisational boundary, disconnected from the internet, thus safeguarding your data. This vision is not just feasible but increasingly necessary in a landscape where data privacy and security are paramount. Open source LLMs offer a tangible solution to these challenges, providing the benefits of advanced AI applications while ensuring data remains within the protective perimeter of the organisation.

Misuse and ethical dilemmas

AI's capability to generate convincing fake content or to be used in scams, such as voice cloning scams, poses significant ethical and security risks. These technologies have been used to fraudulently impersonate individuals, demonstrating the consequences of AI

misuse. Furthermore, instances of AI involvement in creating ransomware or legal disputes over practising law without a licence illustrate the legal and ethical quagmires arising from deploying AI without adequate oversight.

We can take control of the guardrails to be put at the top of the OS LLMs we have deployed.

- ***Unauthorised data scraping:*** This involves collecting data from various sources without permission, which can lead to privacy violations. For instance, there have been cases where large datasets were used to train AI models without the explicit consent of the individuals whose data was included. Imagine a guardrail that sanitises your data that the LLM is consuming.
- ***Creation of deepfakes:*** Deepfake technology uses AI to create highly realistic but entirely fabricated images, videos, or audio recordings. This can be used maliciously to impersonate individuals, spread misinformation, or commit fraud. Imagine a guardrail that takes care of unauthorised creation of such outputs.
- ***AI-generated misinformation:*** An example of this is when AI, like ChatGPT, generates false information, such as incorrectly implicating individuals in legal cases. This unpredictability of AI-generated content can have serious implications, causing harm to reputations or even leading to legal action. Imagine a guardrail that includes citations for every response.

These incidents underscore the complexity and potential risks associated with deploying AI technologies, especially when they are not transparent or open to scrutiny. Proprietary models, which are closed source and controlled by specific organisations, often lack the transparency that allows external parties